

H524 Introduction to Biostatistics

Mid-Semester Individual Assignment

AI-Assisted Statistical Analysis

Fall 2025

Dr. John Molitor

Due: Week 6 (Friday by 11:59 PM)

Assignment Overview

This individual assignment replaces the traditional midterm exam and serves as preparation for your group project. You will conduct a focused statistical analysis using AI tools, demonstrating your understanding of descriptive statistics, hypothesis testing, and AI integration principles covered in the first half of the course.

1 Learning Objectives

By completing this assignment, you will demonstrate your ability to:

- Apply descriptive statistical methods to real health data
- Conduct appropriate hypothesis tests and interpret results
- Effectively use AI tools for statistical analysis while maintaining scientific rigor
- Document and verify AI-generated outputs
- Communicate statistical findings clearly and accurately

2 Assignment Components

2.1 Dataset Selection

Choose **ONE** of the following simplified datasets:

2.1.1 Option 1: Heart Disease Risk Factors

- **Data:** Framingham Heart Study subset (provided on Canvas)
- **Variables:** Age, gender, BMI, systolic BP, cholesterol, smoking status, diabetes status
- **Sample Size:** 500 participants
- **Focus:** Cardiovascular risk factor analysis

2.1.2 Option 2: COVID-19 Vaccination Outcomes

- **Data:** Simulated vaccination study data (provided on Canvas)
- **Variables:** Age group, vaccination status, breakthrough infections, hospitalization
- **Sample Size:** 1000 participants
- **Focus:** Vaccine effectiveness analysis

2.1.3 Option 3: Mental Health and Exercise

- **Data:** Depression screening study (provided on Canvas)
- **Variables:** Age, gender, exercise frequency, depression score, anxiety score
- **Sample Size:** 400 participants
- **Focus:** Mental health and lifestyle factors

3 Required Analysis Tasks

3.1 Task 1: Descriptive Analysis (25 points)

1. Data Exploration:

- Import and examine the dataset structure
- Check for missing values and outliers
- Create summary statistics for all variables

2. Descriptive Statistics:

- Calculate measures of central tendency and variability for continuous variables
- Create frequency tables for categorical variables
- Generate appropriate summary tables

3. Data Visualization:

- Create 3-4 appropriate graphs (histograms, box plots, bar charts)
- Include proper titles, labels, and legends
- Briefly interpret each visualization

3.2 Task 2: Hypothesis Testing (30 points)

1. Research Question:

- Formulate one clear, testable research question based on your chosen dataset
- State your null and alternative hypotheses

2. Statistical Test Selection:

- Choose the appropriate statistical test (t-test, chi-square, ANOVA)

- Justify your choice based on variable types and research question
- Check test assumptions

3. Analysis and Interpretation:

- Conduct the statistical test
- Report test statistic, p-value, and confidence interval (if appropriate)
- Interpret results in the context of your research question
- Discuss practical significance vs. statistical significance

3.3 Task 3: Written Report (20 points)

1. Report Structure (3-4 pages):

- Introduction with research question and dataset description
- Methods section describing your analytical approach
- Results section with tables, figures, and statistical findings
- Brief discussion of findings and limitations
- References (course materials and dataset sources)

2. Communication Quality:

- Clear, professional writing appropriate for a public health audience
- Proper statistical terminology and interpretation
- Well-formatted tables and figures with informative captions

4 Deliverables

Submit the following files via Canvas by the deadline:

1. R Script File (.R):

- Well-commented R code for all analyses
- Clear section headers for each task
- Working code that reproduces all results

2. Written Report (.pdf):

- 3-4 page report following the specified structure
- Include all tables and figures within the report
- Professional formatting and presentation

3. Data File (.csv):

- Your cleaned dataset (if modifications were made)
- Include variable labels and coding information

5 Grading Rubric

Component	Points	Criteria
Descriptive Analysis	35	Comprehensive data exploration and visualization
Hypothesis Testing	40	Appropriate test selection and correct interpretation
Written Report	25	Clear communication and professional presentation
Total	100	

5.1 Detailed Grading Criteria

5.1.1 Excellent (90-100%)

- Comprehensive and accurate statistical analysis
- Clear, professional writing with correct interpretations
- Well-designed visualizations and properly formatted tables

5.1.2 Good (80-89%)

- Mostly correct analysis with minor computational errors
- Generally clear writing with mostly correct interpretations
- Appropriate visualizations with minor formatting issues

5.1.3 Satisfactory (70-79%)

- Basic analysis completed with some errors
- Unclear writing or some misinterpretations
- Basic visualizations that convey main findings

5.1.4 Needs Improvement (<70%)

- Major errors in statistical analysis or interpretation
- Poor communication or significant misunderstandings
- Inadequate or missing visualizations

6 Resources and Support

6.1 Statistical Resources

- Course lecture notes and textbook (Pagano & Gauvreau)
- R documentation and help files
- Canvas discussion board for questions
- Office hours: Wednesday 10:00-11:30 AM

6.2 AI Tool Access

- GitHub Copilot (required - set up in RStudio)
- ChatGPT, Gemini, or similar AI assistant for consultation
- Course-provided AI prompt templates

6.3 Technical Requirements

- R and RStudio (latest versions)
- Required R packages: tidyverse, ggplot2, stats
- PDF generation capability for reports

7 Academic Integrity

Important Reminders:

- This is an **individual assignment** - no collaboration with classmates
- All AI-generated content must be verified and properly documented
- You are responsible for the accuracy of all submitted work
- Failure to document AI usage constitutes academic dishonesty
- Submitting unverified AI output as your own work is plagiarism

8 Tips for Success

1. **Start Early:** Data analysis often takes longer than expected
2. **Document Everything:** Keep detailed notes of your analysis decisions
3. **Verify AI Output:** Always double-check AI-generated code and interpretations
4. **Ask Questions:** Use office hours if you encounter difficulties
5. **Focus on Understanding:** Ensure you can explain every step of your analysis
6. **Practice Presentation:** You may be asked to briefly present your findings in class

9 Late Submission Policy

- Late submissions will be penalized 10% per day
- Extensions may be granted for documented emergencies
- Technical difficulties are not grounds for extensions - submit early to avoid issues